

Sampling and Representation

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can define a question and target population.

I can describe how a sample was selected.

I can limit conclusions to supported groups.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Overfitting Intuition. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 19: Sampling and Representation

A sample must reflect the people and situations where the system will be used.



THINKER MISSION

Audit a playground survey collected from one class.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

If 6 of 10 responses choose Reading, what is the observed proportion?

Readiness for Graphs and honest scales

Counts are 8 and 10. What is their absolute difference?

My readiness explanation

This week’s end goal

By the end, I can define a population, selection frame and nonresponse report for a survey.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use sampling and representation to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A fictional school wants to know which activity Grade 7 students prefer. Surveying only the coding club is easy, but the question concerns every Grade 7 student. First define the population, selection process, question and response options. No real student data is required for this exercise.

Apply the connected idea: Graphs and honest scales

A fictional poll records 8 votes for Drawing and 10 for Building. A bar chart should make lengths proportional to counts, starting at zero. A chart starting at 7 makes the visible bars 1 and 3 units tall and can exaggerate the difference.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



IN

INPUT

Audit a playground survey collected from one class.



PROCESS

A sample must reflect the people and situations where the system will be used.





OUT

OUTPUT

Audit a playground survey collected from one class.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
rows = [{"group": "club", "choice": "Code"},
        {"group": "club", "choice": "Read"},
        {"group": "club", "choice": "Code"}]

code_votes = 0

for row in rows:
    if row["choice"] == "Code":
        code_votes = code_votes + 1

print(code_votes, len(rows))
```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: Target

Evidence: All Grade 7 students

Meaning: Population

Step or case: Available data

Evidence: Three club responses

Meaning: Sample

Step or case: Observed result

Evidence: 2 of 3 choose Code

Meaning: Describes respondents

Read each row from left to right and connect the evidence to the stated rule.

The result 2 of 3 describes these fictional club responses. It cannot by itself estimate the whole grade reliably. A better plan starts from a complete grade roster, uses a fair selection method and records nonresponse. Random selection gives each listed student a chance; it does not guarantee every small sample is representative.

Match the people to the question

The population is everyone the question concerns. The sample is the smaller group whose responses you actually collect. If the question is about all Grade 7 students, a coding-club sample leaves other students out before the poll even starts. Separate the invitation process from the response process: some invited students may not reply. A report needs both counts so readers can see whose preferences might be missing.

Apply the connected idea: Graphs and honest scales

```
counts = {"Drawing": 8, "Building": 10}

for label in counts:

    print(label, "#" * counts[label], counts[label])
```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: Drawing

Evidence: #####

Meaning: 8 votes

Step or case: Building

Evidence: #####

Meaning: 10 votes

Read each row from left to right and connect the evidence to the stated rule.

The text chart uses one # per vote and a zero baseline. The second category has 2 more votes and 1.25 times as many votes, not three times as many. Include the poll question, date or collection occasion and number of responses when presenting a real graph.

Make bar lengths mean something

With a zero baseline, eight votes make a bar eight units high and ten votes make a bar ten units high. The difference is two votes. If the drawing begins at seven, only the extra one and three units are visible, making the second bar look three times as tall. That picture describes the part above seven, not the full counts. Use one scale, label the units and show the counts.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected everyday case



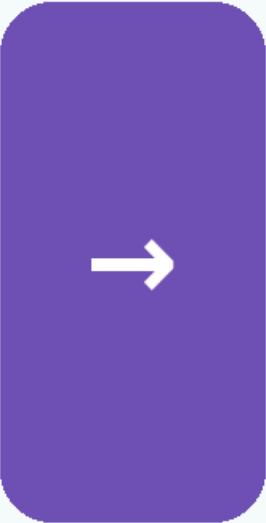
BOUNDARY
Near a decision edge

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



TRANSFER

Fresh situation or user

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Invite only the 15 library-club members instead.

Prediction: The invitation process no longer covers the whole grade through this frame.

Reason: Club members may differ in reading interests; selection is now limited to one group.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

SAMPLE

smaller group used to learn
about a population

2

POPULATION

whole group of interest

3

REPRESENTATION

coverage of relevant groups
and situations

4

SAMPLING BIAS

collection method
systematically misses parts

Word · My own example · Not this / boundary

Word: sample

My own example: A selected group of observations, or an outcome drawn from a distribution; the context identifies which meaning applies.

Not this / boundary: _____

Word: population

My own example: The full group of cases or people a study aims to describe.

Not this / boundary: _____

Word: representation

My own example: A way to encode information so a person or system can work with it.

Not this / boundary: _____

Word: sampling bias

My own example: A systematic mismatch between selected examples and the population the study intends to describe.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Sampling frame

Meaning: The list or source from which a sample is selected

Word: Response

Meaning: A recorded answer from a selected participant

Terms for Graphs and honest scales

Term · Meaning

Term: Axis

Meaning: A reference line with a scale on a graph

Term: Baseline

Meaning: The starting reference value on a scale

Term: Bar chart

Meaning: A graph using bar lengths to compare categories

Term: Scale

Meaning: The mapping from values to positions or lengths

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 19 • WORKED EVIDENCE

- 1 Goal: books for the whole school.
- 2 Sample: one advanced-reading club at lunchtime.
- 3 Missing: younger readers, other interests, absent students, different access needs.
- 4 Repair collection across grades, times, interests, and access needs; then recheck.

1. Goal: books for the whole school.
2. Sample: one advanced-reading club at lunchtime.
3. Missing: younger readers, other interests, absent students, different access needs.
4. Repair collection across grades, times, interests, and access needs; then recheck.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

If the school adds 20 more club responses, which coverage problem remains?

Apply the connected idea: Graphs and honest scales

Draw bars for counts 4 and 6 using one square per vote. State the difference and ratio.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Sampling and Representation

A fictional Grade 7 class wants to understand book-format preferences across its whole grade.

The facts we will use

Population: all 60 Grade 7 learners. Frame: current school-approved list of those 60, used only by the teacher.

Teacher randomly invites 15; 12 answer the same neutral comic/prose/either question. No names enter the analysis table.

Sampling and Representation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Define target

The question concerns all Grade 7, not only library-club members.



2. Select invitations

Teacher uses the complete approved list and documented random selection.



3. Record responses

12 replies from 15 invited; three nonresponses remain visible.



4. Report scope

Invitation response rate $12/15=80\%$; preference counts describe respondents.

CHANGE ONE THING

Invite only the 15 library-club members instead.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Define target

What happens and why: The question concerns all Grade 7, not only library-club members.

Step: 2. Select invitations

What happens and why: Teacher uses the complete approved list and documented random selection.

Step: 3. Record responses

What happens and why: 12 replies from 15 invited; three nonresponses remain visible.

Step: 4. Report scope

What happens and why: Invitation response rate $12/15=80\%$; preference counts describe respondents.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A sample plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined. Use a neutral question and preserve nonresponse coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

What this example does not establish

Even the random-invitation plan does not remove nonresponse bias or justify revealing identifiable school lists.

Read the labelled picture: Can 80% response be reported as 80% of the whole grade responding?

Change one thing: Invite only the 15 library-club members instead. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: A large sample from one narrow group represents everyone. Strong move: locate evidence, change one cause, and retest.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Selection and response are separate. Even a fair invitation sample can be distorted if only interested students respond. Report numbers invited and responding and explain who may be absent.

Twenty students are selected fairly but only 8 answer. Does fair selection remove the need to report nonresponse?

Apply the connected idea: Graphs and honest scales

Counts alone do not show why students chose an activity or whether nonrespondents would choose differently. Graph choices should match the question: categories suit bars; ordered time observations can suit a line.

Art has the most votes among 18 responses. Does that establish Art is best for learning?

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Changing the report heading from club responses to all students does not fix sampling. Change the collection plan or narrow the claim. Avoid collecting names when anonymous counts are sufficient; a teacher can manage selection without publishing identities.

Apply the connected idea: Graphs and honest scales

A graph with bars beginning at 7 turns counts 8 and 10 into visible lengths 1 and 3. Redraw from zero and label counts directly. A line chart can use a restricted axis for detail if clearly labelled, but it must not disguise its scale.

Practise with less help: Read the sample result

Use workbook W2 for the connected example “Questions populations and samples”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Questions populations and samples. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Graphs and honest scales

Use its named workbook W2. Identify the given inputs, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Explain the picture

Can 80% response be reported as 80% of the whole grade responding?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Compare sample cards with an intended-user checklist, highlight missing settings and users, and design a broader collection plan.

Compare sample cards with an intended-user checklist, highlight missing settings and users, and design a broader collection plan.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER
1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or evidence path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

Test reader profiles across reading levels, genres, ages, formats, and accessibility needs.

THIS WEEK'S CHECKPOINT Test reader profiles across reading levels, genres, ages, formats, and accessibility needs.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Document whose requests inform the recommender and avoid treating an easily reached group as all intended users.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Audit a playground survey collected from one class.

CHALLENGE Audit a playground survey collected from one class.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 19 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Only adventure-club members answer a school reading survey.

The question

Explain the sampling problem.

Worked explanation

Membership selects for a relevant interest. Include relevant grades and interests with appropriate data practices.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Who was left out of the survey?

Evidence or stage · Value or result

Evidence or stage: Sample

Value or result: Adventure-club members

Evidence or stage: Intended population

Value or result: School readers

Evidence or stage: Missing perspectives

Value or result: Other interests and relevant grades

Only adventure-club members answer a school reading survey.

Explain the sampling problem.

Membership selects for a relevant interest. Include relevant grades and interests with appropriate data practices.

14. Try a new case independently

Independent case: Plan independently

Design an anonymous sample plan to ask all Grade 7 about three activity choices. State population, sampling frame, selection method, question, and how you will report nonresponse.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 19 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

You survey only the chess club about all students' favourite books. Identify the sampling limitation and one improvement.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Test reader profiles across reading levels, genres, ages, formats, and accessibility needs.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A survey about school reading preferences includes only volunteers from a mystery club. Explain why 100

responses may still fail to represent the school. Propose a better sampling plan using fictional data.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional Grade 7 class wants to understand book-format preferences across its whole grade.

By the end, I can · How I will show it

By the end, I can: Define a question and target population

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Describe how a sample was selected

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Limit conclusions to supported groups

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Define a question and target population

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Grade 7 class wants to understand book-format preferences across its whole grade.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Invitation response rate $12/15=80\%$; preference counts describe respondents.

1. Define target
2. Report scope
3. Record responses
4. Select invitations

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Report scope Invitation response rate $12/15=80\%$; preference counts describe respondents. A sample plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined. Use a neutral question and preserve nonresponse coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

Practice 2: explain the deciding evidence

Changed-case practice: Invite only the 15 library-club members instead. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future input.
- 2. The invitation process no longer covers the whole grade through this frame.
- 3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The invitation process no longer covers the whole grade through this frame. Club members may differ in reading interests; selection is now limited to one group.

Repair a missing building block

Review Missing Data (ai-2/18). Goal: Detect absent and unusable values. Short practice: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?. Return to this week and explain what you changed.

Review Overfitting Intuition (ai-2/16). Goal: Compare training and validation evidence for model capacity. Short practice: Does zero training error justify choosing Complex in the original record?. Return to this

week and explain what you changed.

Review Types of Classification Error (ai-2/13). Goal: Build a labelled confusion matrix from matched reference and prediction pairs. Short practice: Which error causes a sound book to be sent for unnecessary repair review?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.